

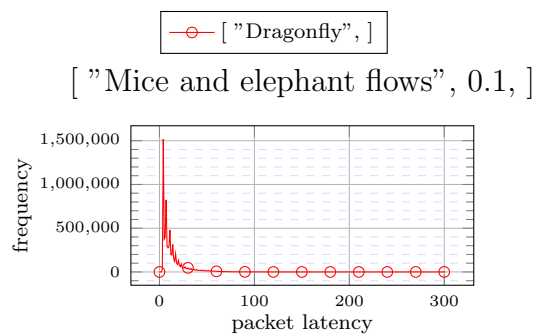


Figure 1: X: ["Mice and elephant flows", 0.1,]

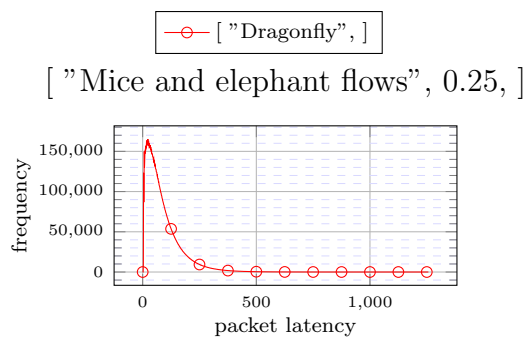


Figure 2: X: ["Mice and elephant flows", 0.25,]

The following versions used in the simulations.

- heads/master-dirty-65e4f7db651d154c0abb3e3d0a691be7d6d92d82(0.6.3)

```
Configuration{
  random_seed: ![ 42, 935, 128643 ],
  warmup: 20000,
  measured: 50000,
  general_frequency_divisor: 2,
  statistics_packet_definitions: [
    [ ],
    [ ]],
  topology: topologia![
    Dragonfly{ global_ports_per_router: 8, servers_per_router: 8, global_arrangement: Palmtree, group_size: 16, number_of_groups: 129, legend_name: "Dragonfly" }],
  traffic: ![
    SyntheticTrafficDistribution{
      tasks: 16512,
      load: ![ 0.1, 0.25 ],
      pattern: Uniform,
      message_size: Multimodal{
        sizes: [ 16, 256 ],
        weights: [ 0.9, 0.1 ]},
      legend_name: "Mice and elephant flows"}],
  maximum_packet_size: 16,
  router: InputOutput{
    virtual_channels: topologia![ 4 ],
    virtual_channel_policies: topologia![
      [
        MapLabel{
          label_to_policy: [
            Chain{
              policies: [
                OccupancyFunction{ label_coefficient: 0, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space:
true, use_neighbour_space: true, aggregate: false }]],
              Chain{
                policies: [
                  AverageOccupancyFunction{
                    label_coefficient: 0,
                    occupancy_coefficient: 2,
                    product_coefficient: 0,
                    constant_coefficient: 16,
                    use_internal_space: true,
                    use_neighbour_space: true,
                    exclude_minimal_ports: true,
                    virtual_channels: [ 0 ]}]},
                  LowestLabel,
                  EnforceFlowControl,
                  Random]],
            allocator: Random,
            buffer_size: 128,
            bubble: false,
            flit_size: 16,
            intransit_priority: false,
            allow_request_busy_port: true,
            output_buffer_size: 64,
            crossbar_frequency_divisor: 1,
            crossbar_delay: 2},
    routing: topologia![
      Sum{
        policy: TryBoth,
        first_routing: ChannelMap{
          routing: AscendantChannelsWithLinkClass{
            routing: WeighedShortest{
              class_weight: [ 1, 100 ]},
            bases: [ 1, 1 ]},
          map: [
            [ 0 ],
            [ 1 ]],
          second_routing: Valiant4Dragonfly{
            first: ChannelMap{
              routing: AscendantChannelsWithLinkClass{
                routing: WeighedShortest{
                  class_weight: [ 1, 100 ]},
                  bases: [ 1, 1 ]},
                map: [
                  [ 0 ],
                  [ 1 ]],
                second: ChannelMap{
                  routing: AscendantChannelsWithLinkClass{
                    routing: WeighedShortest{
                      class_weight: [ 1, 100 ]},
                      bases: [ 1, 1 ]},
                    map: [
                      [ 2 ],
                      [ 3 ]]}},
                  first_allowed_virtual_channels: [ 0, 1, 2, 3 ],
                  second_allowed_virtual_channels: [ 0, 1, 2, 3 ],
                  second_extra_label: 1,
                  legend_name: "ugal - minFirst"}],
            link_classes: [
              LinkClass{ delay: 2 },
              LinkClass{ delay: 2 },
              LinkClass{ delay: 2 }],
            launch_configurations: [
              Slurm{ job_pack_size: 1, time: "3-00:00:00" }]]}
```